

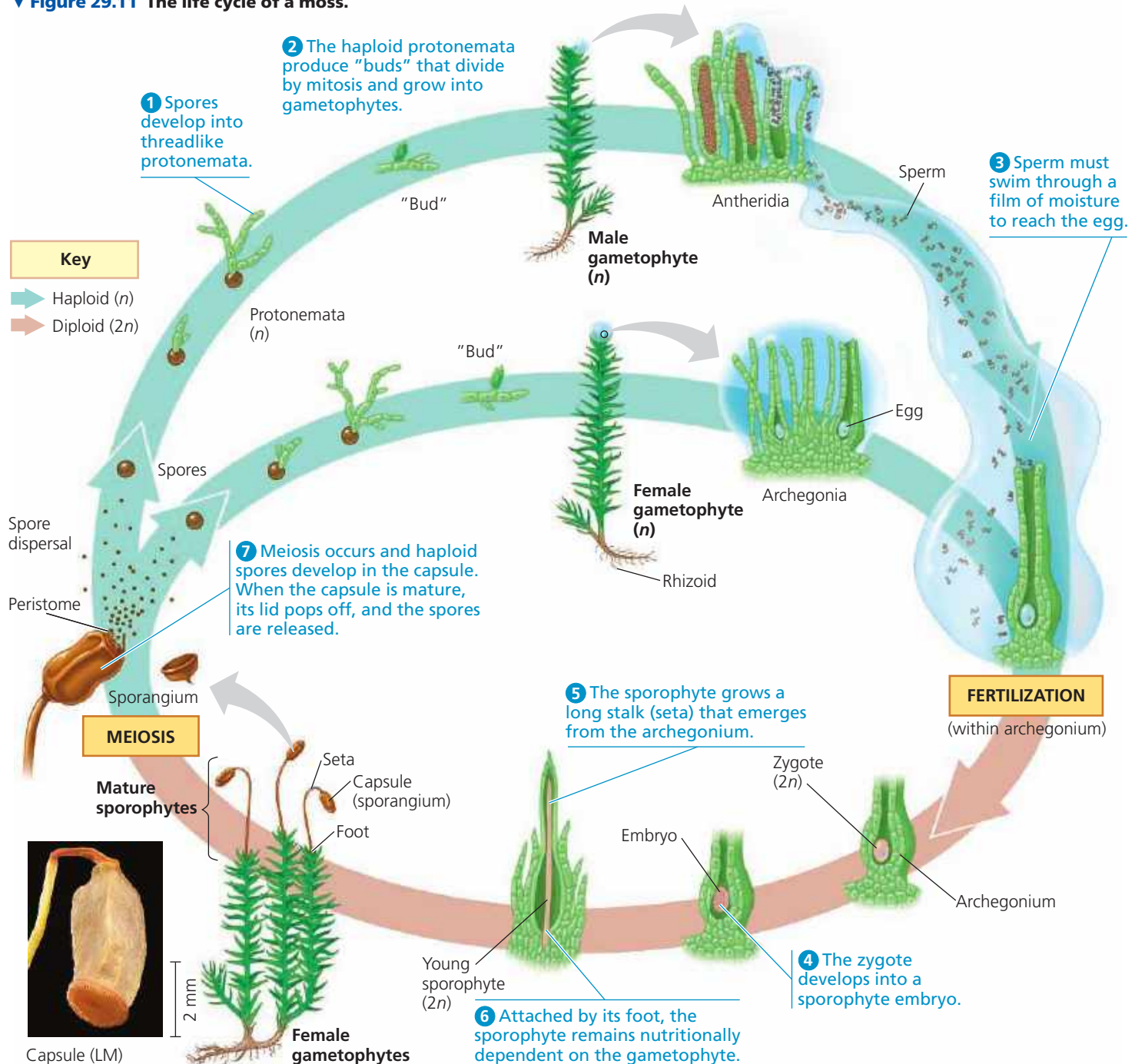
liverworts, and by 430 million years ago spores similar to those of mosses and hornworts also occur in the fossil record. The earliest fossils of vascular plants date to about 425 million years ago.

Over the long course of their evolution, liverworts, mosses, and hornworts have acquired many unique adaptations. Next, we'll examine some of those features.

Bryophyte Gametophytes

Unlike vascular plants, in all three bryophyte phyla the haploid gametophytes are the dominant stage of the life cycle: They are usually larger and longer-living than the sporophytes, as shown in the moss life cycle in **Figure 29.11**. The sporophytes are typically present only part of the time.

▼ **Figure 29.11** The life cycle of a moss.



VISUAL SKILLS In this diagram, does the sperm cell that fertilizes the egg cell differ genetically from the egg? Explain.

➔ **Mastering Biology Animation: Moss Life Cycle**